MILITARY SPECIFICATION

RANGEFINDER SYSTEM, FIRE CONTROL: LASER AN/GVS-3

WARNING - EYE HAZARD

EXTREME CAUTION MUST BE EXERCISED IN OPERATING AND TESTING THE RANGEFINDER. IF THE LASER LIGHT BEAM ENTERS THE HUMAN EYE, THE RETINA OF THE EYE CAN BE BURNED. THE HIGH ENERGY DENSITY OF THE BEAM CAN ALSO PRODUCE SUCH DAMAGE AT EXTREME RANGE BY REFLECTION FROM ANY SHINY SURFACE SUCH AS GLASS, METAL, PAINT, ETC. IN VIEW OF THE POSSIBILITY OF PRODUCING SERIOUS EYE DAMAGE, THE RANGEFINDER SHOULD NOT BE OPERATED UNLESS THE TARGET AREA HAS BEEN EVACUATED.

1. SCOPE

1.1 This specification covers one type of portable rangefinder system for use by the Artillery forward observer.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel, General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment and Tools for Army Material: Packaging and Packing of
MIL-I-45607	Inspection Equipment, Acquisition, Maintenance and Disposition of
MIL-C-60260	Chest
MIL-C-60250	Carrying Cases
MIL-A-60257	Adapter Mount
MIL-B-60298	Battery, Nickel Cadmium, Vented: 10541198
MIL-C-60299	Charger, Battery: 10541243
MIL-T-60500	Transceiver

FSC 1240

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STANDARDS

Military

MIL-STD-105	Sampling, Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-810	Environmental Test Methods

DRAWINGS

U.S. Army, Frankford Arsenal

A10541244	Rangefinder System, Fire Control, Laser: AN/GVS-3
F10541045	Chest
F10541038	Case, Carrying
F10540501	Transceiver
F10541026	Case, Carrying
F10541078	Tripod
F10541111	Mount, Adapter
F10541103	Mount, Shelf
D10541240	Chart, Boresight
A10541242	Screwdriver
F10541243	Charger, Battery
F10541198	Battery Pack
D10541241	Holder, Cable
C10541236	Cable, 50 ft.
C10541237	Cable, 12 ft.
D10544852	Cable, 7 ft.

PACKAGING DATA SHEET

10541244	Rangefinder System, Fire Control, Laser AN/GVS-3

(Copies of specifications, standards, drawings and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Materials.- Materials shall be in accordance with the drawings, material specifications and general specifications forming a part of this specification.

3.2 Fabrication.- The system shall be manufactured in accordance with Drawing A10541244 and drawings pertaining thereto. The following assemblies shall each have met their detailed specifications prior to assembly as a system into Configurations A through J, defined in 3.4, Table II.

a. Transceiver: 10540501 - MIL-T-60500
b. Adapter Mount: 10541111 - MIL-A-60257
c. Battery Charger: 10541243 - MIL-C-60299
d. Battery Pack: 10541198 - MIL-B-60298
e. Carrying Cases: 10541026 & 10541038 - MIL-C-60250
f. Chest: 10541045 - MIL-C-60260

3.3 General specification.- The system shall meet the following requirements, where applicable, of MIL-F-13926:

a. Dimensions and tolerances
b. Inorganic protective surface finishes
c. Electrical and electronic assemblies
d. Part identification and marking
e. Workmanship

3.4 Configuration.- The operating, carrying, installation and storage configurations listed in Table II consist of the items listed in Table I when connected for operation, carried, stored or installed in accordance with the corresponding procedures of paragraph 6.3.

TABLE I - ITEM LIST

ITEM	PART NUMBER	ITEM LIST
Transceiver	10540501	1
Mount, Adapter	10541111	2
Tripod	10541078	3
Battery Pack	10541198	4
Charger, Battery	10541243	5
Case, Carrying (R.F.)	10541038	6
Case, Carrying (Acc.)	10541026	7
Chest	10541045	8
Screwdriver	10541242	9
Chart, Boresight	10541240	10
Technical Manual	TM 9-1240-316-14	11
Shelf Mount	10541103	12
Cable, 12 ft.	10541237	13

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Cable, 7 ft.	10544852	14
Cable, 50 ft.	10541236	15
Cable Holder	10541241	16
Tripod, Universal	FSN 6675-641-3572	17

TABLE II - SYSTEM CONFIGURATION LIST

CONFIGURATION	ITEMS INCLUDED AS IDENTIFIED IN TABLE I
A. (Battery operation of R.F. on Tripod)	1, 2, 3, 4
B. (Battery operation of R.F. on shelf mt)	1, 2, 4, 12
C. (Charger operation of R.F. on Tripod 400 cps A.C.)	1, 2, 3, 5, 13, 15
D. (Charger operation of R.F. on Tripod 60 cps A.C.)	1, 2, 3, 5, 13, 15
E. (Operation from jeep)	1, 2, 3, 5, 14, 15
F. (Back Pack)	1, 2, 3, 4, 6, 7, 12
G. (Charger w/batteries, cover open)	4, 5
H. (Chest Loaded w/cover open)	1, 2, 3, 4, 6, 7,8, 9, 12
I. (Field trans. and storage)	Items 1 through 16
J. (Battery operation of R.F. on universal tripod)	1, 2, 17, 4

3.5 Environmental.- The system shall meet the following environmental requirements in the configurations specified.

3.5.1 Vibration.- The system while thermally stabilized at -65 ± 2°F in Configuration I shall withstand vibration for 15 minutes each in 3 mutually perpendicular directions. Vibration in each direction shall be at frequencies varied logarithmically from 5 to 500 cycles per second. The double amplitude of applied vibration shall be 0.036 inch at lower frequencies and shall be maintained at, but not exceed, a 5g level at higher frequencies. Subsequent to vibration, the system shall show no evidence of damage and Configurations A through E of the system subjected to vibration shall meet the applicable requirements of 3.5.12 through 3.10.3.

3.5.2 Shock.- The system in Configuration I shall withstand a total of 18 shock impulses, 3 in each direction of 3 mutually perpendicular axes. Each shock impulse shall be a half-sine wave with a time duration of 11 ± 1 milliseconds. The peak amplitude of each shock impulse shall be 75g's. Subsequent to shock, the system shall show no evidence of damage and Configurations A through H of the system subjected to shock shall meet the applicable requirements of 3.5.12 through 3.10.3.

3.5.3 Rough handling.- The system in Configuration F shall withstand rough handling as simulated by Part 2, Procedure XI of MIL-STD-810 while thermally stabilized at +75°F. Transverse motion shall not exceed 3 inches.

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Subsequent to rough handling, the system shall show no evidence of damage and shall meet the requirements of 3.7.4 as well as the applicable requirements of 3.5.12 thru 3.10.3 for Configuration A and B of the system subjected to rough handling.

3.5.4 Rain.- The system in Configurations A through H shall each be capable of withstanding 5 hours of continuous exposure to heavy rainfall in the amount of 2.0 inches/hour with a wind speed of 35 knots at a water temperature of plus 85°F. Subsequent to such rainfall, Item 1 shall be subjected to uncovered storage at +50° ± 5°F for 2 hours or more immediately followed by storage at +75° ± 5°F for 1 hour. There shall be no evidence of water penetration inside of Item 1 at any time and the system shall not have been damaged or adversely affected and shall meet the applicable requirements of paragraphs 3.5.12 through 3.10.3 for Configurations A through E of the system exposed to rain.

3.5.5 Submersion.- The system in Configuration I, thermally stabilized at +90 ± 5°F, shall withstand submersion under 3 ft. of water at a temperature of 70 ± 5°F for at least 15 minutes. Subsequent to submersion, the system shall have no evidence of water penetration inside the volume of Items 5, 8, 14, and 15. The system shall not have been damaged or adversely affected and in Configurations A through E shall meet the applicable requirements of 3.5.12 through 3.10.3.

3.5.6 Humidity.- The system in Configurations A through H shall be capable of withstanding 4 hours of continuous exposure to a relative humidity of 90 to 95 percent at a temperature of plus 40° ± 2°C. During and subsequent to exposure to humidity, the system shall not have been damaged or adversely affected, and in Configurations A thru E shall meet the applicable requirements of 3.5.12 through 3.10.3.

3.5.7 Sand and dust.- The system in Configurations A through I shall be capable of withstanding 3 hours of continuous exposure to blowing sand of particle size 0.01 to 1.00mm and blowing dust of particle size 0.001 to 0.01mm with a density between 0.1 to 0.5 gram per cubic foot of sand and a density between 0.1 to 0.5 gram per cubic foot of dust at a temperature of plus 25°C, a relative humidity between 15 to 30%, and a minimum wind velocity of 15 knots. Subsequent to sand and dust exposure, the system shall not have been damaged or adversely affected and in Configurations A through E shall meet the applicable requirements of 3.5.12 through 3.10.3.

3.5.8 Salt spray.- The system in Configurations A through I shall be capable of withstanding 48 hours of continuous exposure to a fine spray of 5% salt solution concentration with a specific gravity of from 1.0268 to 1.0413, a PH between 6.5 and 7.2, and a temperature between plus 93°F and plus 97°F. Subsequent to salt spray, the system shall not have been damaged or adversely affected and in Configurations A through E shall meet the applicable requirements of 3.5.12 through 3.10.3.

3.5.9 Altitude.

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3.5.9.1 Operating.- The system in Configurations A through E shall be capable of operating during 4 hours of continuous exposure to an altitude of 10,000 feet at a temperature of +75°F ± 15°F. During and subsequent to such exposure, the system shall not have been damaged or adversely affected and shall meet the applicable requirements of 3.5.12 through 3.10.3.

3.5.9.2 Non-operating.- The system in Configurations F, G and I shall be capable of 4 hours of continuous exposure to an altitude of 50,000 feet at a temperature of -65°F ± 5°F. Subsequent to such exposure, the system shall not have been damaged or adversely affected and in Configurations A through E, shall meet the requirements of 3.5.12 through 3.10.3.

3.5.10 Fungus.- The system in Configuration I shall be capable of 28 days exposure to a composite spore suspension consisting of the following cultures: (a) Chaetomium globosum, (b) Aspergillus versicalor, (c) Aspergillus flavus, (d) Pencillium luteum, (e) Rhizopus nigricans. The composite spore suspension shall be incubated at an ambient temperature of between +77°F and +84°F with an RH of 97 ± 2%. Subsequent to the above fungus exposure, the system shall not have been damaged or adversely affected and in Configurations A through E shall meet the requirements of 3.5.12 through 3.10.3.

3.5.11 Storage temperatures.- The system in Configuration I shall be capable of withstanding temperature stabilization at +160°F and -80°F for a period of 8 hours at each temperature extreme. Subsequent to exposure to storage temperatures, the system shall not have been damaged or adversely affected and Configurations A through E of the system exposed to storage temperatures shall meet the requirements of 3.5.12 through 3.10.3.

3.5.12 Operating temperature.

a. Unless otherwise specified the system in Configurations A and B shall meet the applicable requirements of 3.7 through 3.10.3 while thermally stabilized at temperatures from -25°F to 155°F inclusive.

b. Unless otherwise specified the system in Configurations C, D and E shall meet the requirements of 3.7 through 3.10.3 while thermally stabilized at temperatures from -65°F to 155°F inclusive.

3.6 Reliability.- The system shall exhibit a mean rangings between failure (MRBF) of 1500 rangings at a 95% confidence level.

3.7 Performance.- The system configured into Configuration A through E shall meet the following requirements as they apply to specific configurations:

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3.7.1 Target location.- The system in Configurations A through E shall meet the following requirements:

3.7.1.1 Ranging.- The system shall indicate the range to targets, located from 230 meters and 20,000 meters inclusive, where atmosphere and range reduce energy density emitted from the rangefinder by a factor of 10^{10} or less.

3.7.1.1.1 Ranging accuracy.- Range measured to targets from 230 meters to 20,000 meters shall be accurate to ± 10 meters.

3.7.1.1.2 Range resolution.- The system shall distinguish between targets separated by 30 meters in range and shall indicate the range to the first target, then the range to the second target on successive rangings.

3.7.1.1.3 Target discrimination.- Upon firing, the system shall indicate the presence of one, two, three, four and five or more targets visible within the boresight square of the telescope reticle when these targets are separated by a distance of 20 to 45 meters and are beyond the selected setting of the minimum range switch. The system shall also indicate the range to the target corresponding to the selected position of the target discriminator switch upon the same firing. The indication of the number of targets shall be displayed by the lighting of the appropriate target discriminator lamp when the read switch is depressed.

3.7.1.1.4 Minimum range selection.- With the minimum range selector switch rotated to each of the positions indicated in Column I, the system, when fired at a target at the range indicated in Column II, shall result in a "no target" indication when the read switch is depressed. When the system is fired at a target at the range indicated in Column III, the system shall indicate the range to the accuracy specified in 3.7.1.1.1.

I Selector Position	II Range	III Range
.2 = 200 meter	185 meter	225 meter
.6 = 600 meter	585 meter	625 meter
1 = 1000 "	985 "	1025 "
2 = 2000 "	1985 "	2025 "
3 = 3000 "	2985 "	3025 "

3.7.1.1.5 Rapid ranging.- The system shall indicate the range to a target located at 1500 meters each time it is fired when fired at 15 second intervals for 5 consecutive firings, followed by 10 consecutive firings at 1 minute intervals.

3.7.1.1.6 Delay ranging.- The system shall indicate the correct range to a target at approximately 5000 meters to within the range

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accuracy of 3.7.1.1.1 when the range switch is depressed no less than 75 seconds after the charge switch has been depressed at RT only.

3.7.1.1.7 Minimum time between rangings.- The system shall be capable of cycling through a charging and firing sequence within 10 seconds.

3.7.1.1.8 Self-ranging.- The system shall range only when the range switch is depressed.

3.7.1.2 Electronic display.

3.7.1.2.1 Turn-on display.- With the power switch initially placed in the "ON" position and read switch depressed, the system shall read four zeros at the nixie tubes with the number seven superimposed on the zero in the unit column and no target indicated (no light) on the five neon target discriminators.

3.7.1.2.2 Time interval counter test.- With the read switch depressed subsequent to depressing the test switch, the counter shall indicate a reading on the numerical indicators of between twenty and fifty meters.

3.7.1.2.3 Range counter reset.- The counter shall be reset by placing the power switch in the ON position and also when the RANGE switch is depressed ten seconds or more after the CHARGE switch has been depressed. Indication that the counter is reset shall be given by four zeros with the number seven superimposed on the zero in the unit column of the numerical readout when the READ switch is depressed.

3.7.1.2.4 Long range reading.- When range is read on the numerical indicators, the number one shall be superimposed on the number that appears in the units column, when ranging to a target in excess of 9999.

EXAMPLE - Numbers that appear on the four nixie tubes with a "0" in the units column, and are known to be targets beyond 1000 meters, indicate a target in the 10,000mm range e.g. 0735 indicates a range reading of 10,735 meters.

3.7.1.2.5 Target discriminator lamps.

a. No more than one target discriminator lamp shall light to indicate the presence of multiple targets.

b. The one to five neon target discriminator lamps shall light and extinguish in sequence at an average rate of once per second when setting the receiver noise level.

3.7.1.2.6 Low voltage indicator.- With the power switch in the ON position and the READ switch depressed the numeral two shall appear in the unit column in addition to Numbers 0, 1, 5 or 7 whenever the input

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voltages are less than 21 ± 1 volts or less than 7.2 ± 0.2 VDC.

3.7.1.3 Elevation.

3.7.1.3.1 Illumination.- Depression of the switch on the yoke assembly shall illuminate the elevation and azimuth scales, and a circular level vial on the adapter mount. The intensity of the reticle light shall be smoothly variable from maximum brightness to completely off by turning the reticle adjustment potentiometer knob on the cover plate of the telescope assembly in the counterclockwise direction.

3.7.1.3.2 Elevation accuracy.- With the horizontal reticle line in coincidence with a horizontal target line, and the elevation scale and micrometer each in coincidence with their respective index, the scale and micrometer shall indicate the true position of the line of sight in elevation and depression including backlash, within 2.0 mil at elevation and depression readings of 75, 150, 300, and 500 mils.

3.7.1.3.3 Elevation backlash.- Backlash in the elevation mechanism at any point over the full excursion of travel 500 mils elevation and 500 mils depression from a horizontal line of sight shall not exceed 0.8 mil.

3.7.1.3.4 Elevation knob torque.- The running torque required to rotate the elevation knob in each direction of clockwise and counterclockwise rotation of the elevation knob shall not exceed the values specified at temperatures indicated.

-65°F	counterclockwise	5 inch pounds
	clockwise	6.5 inch pounds
70° to 80°F	counterclockwise	25 inch ounces
	clockwise	40 inch ounces
+155°F	counterclockwise	25 inch ounces
	clockwise	40 inch ounces

3.7.1.3.5 Lateral displacement.- Any horizontal movement of the line of sight caused by reversing direction of rotation of the elevation knob shall not exceed 0.10 mil at any point throughout the excursion of plus and minus 500 mils.

3.7.1.4 Azimuth.

3.7.1.4.1 Azimuth excursion.- The azimuth excursion through which the adapter mount shall move a transceiver, secured to the adapter mount, shall be at least 5675 mils.

3.7.1.4.2 Azimuth error.- The error in azimuth scale readings taken in 10 mil steps throughout the total excursion (6400 mils) of the azimuth scale shall not exceed 2.0 mils. The total spread between the largest positive error and the largest negative error shall not exceed 4.0 mils.

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3.7.1.4.3 Azimuth knob rotation.- The maximum torque required to rotate the azimuth adjustment knob for azimuth rotation of a transceiver shall not exceed 30 inch-ounces when thermally stabilized at room temperature and shall not exceed 60 inch-ounces when the adapter mount with transceiver is thermally stabilized at the extreme temperature requirements of 3.6.12.b.

3.7.1.4.4 Microscope.- The azimuth scale of the adapter mount as read through the microscope shall be clear and in sharp focus. The difference in focus between the image of the reticle in the microscope and the azimuth scale shall not exceed 0.5 diopter.

3.7.1.5 Leveling.

3.7.1.5.1 Circular level.- The level bubble of the adapter mount shall remain within the inner circle of the level, when a transceiver is moved in azimuth through an excursion of 5675 mils.

3.7.1.5.2 Plumb travel.- With the system leveled as in 3.7.1.5.1 and the elevation scale of the system indicating 500 mils elevation, the line of sight through the center of the boresight square of the reticle shall be brought into coincidence with a plumb line. The line of sight through the reticle shall track the plumb line within plus or minus 1.0 mil (total excursion of deviation not to exceed 1.0 mil) when the system line of sight is depressed over an excursion of 1000 mils by rotation of the elevation knob. This requirement shall be met without need for releveling when the transceiver is rotated 90° in azimuth (approximately 1600 mils).

3.7.1.6 Sighting.

3.7.1.6.1 Open sight.- The line of sight through the open sight shall be parallel within 20 mils in azimuth and elevation to the established line of sight of the sighting optical system.

3.7.1.6.2 Collimation (boresight).- Using Chart, Boresight, 10541240 made in accordance with Drawing D10541240, positioned on a horizontal line of sight at a known distance of between 100 to 150 feet from the rangefinder, the rangefinder shall be oriented and fired with parallax shield closed to position the laser beam in the center of the rectangle on the target. With the rangefinder at this position, through boresight adjustments, the horizontal and vertical reticle lines shall be positioned in coincidence with the horizontal and vertical target lines. The elevation scale on the telescope assembly and the dial scales of the micrometer shall each be adjusted to coincide with their respective zero indexes and locked in position.

3.7.1.6.3 Boresight adjustment.- The maximum torque required for rotation of the azimuth and elevation boresight screws shall not exceed 3 inch-pounds when the system is thermally stabilized at 75° ± 2°F and shall not exceed 5 inch-pounds when thermally stabilized at the temperature extremes of 3.5.12b.

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3.7.1.6.4 Eyepiece focus.- The diopter scale shall indicate zero diopters within plus or minus 1/4 diopter when the eyepiece focus is adjusted to zero diopters.

3.7.1.6.5 Diopter adjustment.- The maximum torque required for rotation of the diopter adjustment for eyepiece focus shall not exceed 10 inch-ounces when the system is thermally stabilized at 75° ± 5°F and shall not exceed 15 inch-ounces when thermally stabilized at the temperature extremes of 3.5.12b.

3.7.1.6.6 Reticle illumination.- With the reticle lamp energized with 28 volts the reticle pattern shall be clearly defined when viewed in a darkened area. Rotation of the potentiometer knob clockwise from the extreme counterclockwise position shall complete the electrical circuit and then increase the intensity of reticle illumination from minimum to maximum.

3.7.2 Power sources.

3.7.2.1 Battery power.- At temperatures from -25° to +155°F inclusive, the system in Configurations A and B shall be capable of meeting the requirements of 3.7.1 through 3.7.1.6.6 for 100 rangings with the same battery pack without recharging.

3.7.2.2 Charger function.

a. The system in Configurations C, D and E shall be capable of meeting the requirements of 3.7.1 through 3.7.1.6.6 at temperatures from -65°F to +155°F.

b. At temperatures from -25°F to +155°F, the system in Configurations C or D, Item 5, shall be capable of charging one to three completely discharged battery packs to full charge in 5 hours or less while the system simultaneously meets the requirements of 3.7.1 through 3.7.1.6.6. Subsequent to charging the system shall meet 3.7.2.1.

3.7.3 Mounting stability.- The system in Configurations A through E and Configuration J shall be capable of maintaining a stable configuration while meeting the requirements of 3.7.1 through 3.7.1.6.6. The tripod and shelf mount of the systems shall accept and securely lock the system and shall meet the requirements of 3.7.3.1 and 3.7.3.2.

3.7.3.1 Tripod.- The legs of the tripod (Item 3) shall slide without interference when the tripod is adjusted from minimum to maximum height. With the legs of the tripod extended to maximum height, the brakes of the tripod legs shall maintain the system at maximum height and maintain system level during system operation.

3.7.3.2 Shelf mount.- The stakes of the shelf mount (Item 12) shall be capable of being inserted into the angular hole in the base such that

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the flange of the stake is seated against the face of the angular hole.

3.7.4 Portability.- The system in Configuration F shall meet the requirements of 3.7.4.1 and 3.7.4.2.

3.7.4.1 Weight.- The weight of individual items of the system shall not exceed the following:

Item	Item No.	Maximum Weight in lbs.
Transceiver	1	20.2
Adapter mount	2	3.0
Tripod	3	2.7
Battery pack	4	5.0
Carrying case	6	2.2
Carrying case	7	2.6
Shelf mount (Including 3 stakes)	12	1.6

3.7.4.2 Operability.- The interlocking fasteners, cable connectors, hinges, shoulder strap buckles, and mechanical strap fasteners shall be manually operable.

3.8 Electromagnetic compatibility.- The radiated and conducted electromagnetic emissions of the system in Configurations A through E shall not exceed the allowable limits of MIL-STD-461 for Class IIA equipment.

3.9 Interchangeability.- The system shall meet all the requirements of this specification without need for selective assembly of Items 1 through 17, excluding Items 9, 10 and 11. They shall be interchangeable such that replacement of any single item or random combination of items of the system permits the system to operate and perform to the requirements of this specification. No incompatibility shall exist between the item replacement and the mechanical interfaces/fits or electrical functions of the system. The following specific interface requirements shall also be met when items are interchanged:

a. Locking torque.- The minimum torque required to rotate the locking knob of the adapter mount (Item 2) through maximum clockwise excursion, to secure a transceiver to the adapter mount shall be 10-12 inch-pounds.

b. Unlocking torque.- The maximum torque required to rotate the locking knob of the adapter mount through maximum excursion, to unlock and release a transceiver from the adapter mount shall not exceed 12 inch-pounds.

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c. Support interface.

(1) The carrying cases of Configuration F and the chest of Configuration I shall interface with and support intended contents.

(2) In Configuration J, the universal tripod shall be capable of seating an adapter mount (with transceiver) and securely fastening it in place. The hook on the battery shall fit on the cross bar of the leg of the tripod and while in this position, the top of the battery shall be capable of being connected to the adapter mount cable connector.

3.10 Safety and equipment protection functions.

3.10.1 Operation and maintenance.

a. When the power switch of the transceiver (Item 2) is in the ON position and the capacitor of the pulse forming network (PFN) is at full charge, placement of the power switch to the OFF position shall cause the PFN to discharge through the flash lamp and the motor shall not run.

b. When the power switch is in the ON position and the range switch is depressed, without depressing the CHARGE switch, the motor shall run but the flash lamp shall not fire.

c. When the capacitors of the PEN are at full charge, removal of the cover of the left housing shall cause the energy stored in the capacitors to discharge through a load resistor.

3.10.2 Eye filter.

a. The absorbing glass filter of the eye filter assembly (10558179) of the transceiver shall attenuate the laser light reflected to the eye by a factor of 100,000.

b. The filter lever of the eye filter assembly shall be operable under all environmental conditions and shall allow for removal of the filter from the line of sight of the transceiver when the lever is depressed and replacement of the filter into its original position into the line of sight of the transceiver when the lever is released.

3.10.3 Equipment protection.

a. The adapter mount shall prevent 360° rotation of the transceiver.

b. The charger shall prevent damage to separate internal functions within the charger.

c. The adapter mount cable shall prevent damage to the transceiver caused by a short circuit within the transceiver.

d. The cable connectors of Items 15, 16 and 17 shall prevent connection to a wrong receptacle within the system.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of 6 rangefinder systems plus 6 each of all items covered by SQAP. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

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4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAP's, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply. Examination, and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables 1, II, III and IV herein. Examination and tests for packaging, packing, and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables 1, II, III and IV shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

TABLE I - CLASSIFICATION OF DEFECTS
(100% INSPECTION)

		REQUIREMENT	TEST PROCEDURE
101.	Ranging	3,7.1.1	4.6.4.1
102.	Ranging accuracy	3.7.1.1.1	4.6.4.2
103.	Range resolution	3.7.1.1.2	4.6.4.3
104.	Target discrimination	3.7.1.1.3	4.6.4.4
105.	Minimum range selection	3.7.1.1.4	4.6.4.5
106.	Rapid ranging	3.7.1.1.5	4.6.4.6
107.	Delay ranging	3.7.1.1.6	4.6.4.7
108.	Minimum time between ranging	3.7.1.1.7	4.6.4.8
109.	Self ranging	3.7.1.1.8	4.6.4.9
110.	Turn-on display	3.7.1.2.1	4.6.4.10
111.	Time interval counter	3.7.1.2.2	4.6.4.11
112.	Range counter reset	3.7.1.2.3	4.6.4.12
113.	Long range reading	3.7.1.2.4	4.6.4.13
114.	Target discriminator lamps	3.7.1.2.5	4.6.4.14
115.	Low voltage indicator	3.7.1.2.6	4.6.4.15
116.	Illumination	3.7.1.3.1	4.6.5.1
117.	Elevation accuracy	3.7.1.3.2	4.6.5.2
118.	Elevation backlash	3.7.1.3.3	4.6.5.3
119.	Elevation knob torque	3.7.1.3.4	4.6.5.4
120.	Lateral displacement	3.7.1.3.5	4.6.5.5
121.	Plumb travel	3.7.1.5.2	4.6.5.6
122.	Azimuth excursion	3.7.1.4.1	4.6.6
123.	Azimuth error	3.7.1.4.2	4.6.6.1
124.	Azimuth knob rotation	3.7.1.4.3	4.6.6.2
125.	Microscope	3.7.1.4.4	4.6.6.3
126.	Circular level	3.7.1.5.1	4.6.6.4
127.	Open sight	3.7.1.6.1	4.6.6.5
128.	Collimation (boresight)	3.7.1.6.2	4.6.6.6

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TABLE I - CLASSIFICATION OF DEFECTS (Cont'd)

		REQUIREMENT	TEST PROCEDURE
129.	Boresight adjustment	3.7.1.6.3	4.6.6.7
130.	Eyepiece focus	3.7.1.6.4	4.6.6.8
131.	Diopter adjustment	3.7.1.6.5	4.6.6.9
132.	Reticle illumination	3.7.1.6.6	4.6.6.10
133.	Battery power	3.7.2.1	4.6.7
134.	Charger function	3.7.2.2(a)(b)	4.6.7.1
135.	Mounting stability	3.7.3	4.6.7.2
136.	Tripod	3.7.3.1	4.6.7.3
137.	Shelf mount	3.7.3.2	4.6.7.4
138.	Weight	3.7.4.1	4.6.7.5
139.	Operability	3.7.4.2	4.6.7.5.1
140.	Interchangeability	3.9	4.6.9
141.	Operation and maintenance	3.10.1	4.6.10
142.	Eye filter	3.10.2(b)	4.6.10.1
143.	Equipment protection	3.10.3	4.6.10.2
144.	Operational temperatures(-65°F,-25°F, 70°F to 80°F, 155°F)	3.5.12	4.6.2.13

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.3.4 Certification.- Certification shall include test data and results for the characteristic specified below. Certification shall be provided prior to performing final acceptance inspection, and remain in effect throughout the contract for Government acceptance. This does not relieve the contractor from inspection throughout the contract production, including recording data and results therefrom. Recorded data and results shall be made available to the Government upon request. A new certification will be required should any changes occur in materials, manufacturing processes and techniques previously certified.

CHARACTERISTIC	TEST DATA TO COMPLY WITH
Eye filter	3.10.2(a)

4.4 Special sampling.

4.4.1 General.

4.4.1.1 Three rnagefinder systems shall be selected at random as a special sample from each 50 produced or from each month's production, whichever occurs first. The sample shall meet the requirements and tests in Table II.

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TABLE II

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301.	Materials	3.1	Applicable drawings & Specs - Visual
302.	Fabrication	3.2	Applicable drawings & Specs - Visual
303.	General specification	3.3	MIL-F-13926 - Visual
304.	Electromagnetic compatibility	3.8	4.6.8

NOTE: The tests of Table II shall be conducted at a temperature between +70° and +80°F.

4.4.1.2 Three rangefinder systems shall be selected at random as a special sample from each month's production. Each sample shall meet the requirements and tests in Table III.

TABLE III

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
305.	Interchangeability	3.9	4.6.9

4.4.2 Environmental.- Three rangefinder systems shall be selected at random as a special sampling from each 50 produced, or from each month's production, whichever occurs first. Each sample shall be subjected to the tests in Table IV and shall then meet the requirements and tests in Table I.

TABLE IV

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
306.	Storage temperatures (-80° and +160°F)	3.5.11	4.6.2.1
307.	Shock (between +70° and 80°F)	3.5.2	4.6.2.2
308.	Vibration (between +70° and 80°F and -65°± 2°F)	3.5.1	4.6.2.3
309.	Rough handling	3.5.3	4.6.2.4
310.	Rain	3.5.4	4.6.2.5
311.	Submersion	3.5.5	4.6.2.6
312.	Humidity	3.5.6	4.6.2.7
313.	Sand and dust	3.5.7	4.6.2.8
314.	Salt spray	3.5.8	4.6.2.9
315.	Altitude-operating	3.5.9.1	4.6.2.10
316.	Altitude non-operating	3.5.9.2	4.6.2.11

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4.4.3 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the represented inspection lot shall be suspended by the Government until necessary corrections have been made by the contractor and resubmitted samples have been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-445607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the List of Documents when provided with the technical data package (TDP).

4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.2.3 Inspection equipment design requirements.

4.5.2.3.1 Performance of rangefinder system.- The systems configurations specified in 6.3 through 6.3.10 shall be utilized to assemble the rangefinder into its various configurations for testing.

4.5.2.3.2 Wall targets.- A calibrated wall target with appropriate target patterns for measuring the optical and mechanical accuracies of the rangefinder system specified in 3.7.1.3 shall conform to the following requirements:

(a) The target shall be positioned approximately 5 feet $\pm$ 1/32 inch in front of the rangefinder system and aligned to the axis of rotation of the

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telescope assembly. The "0" horizontal line of sight of the target image shall be set in the same (leveled) horizontal plane as the "0" horizontal line of sight of the transceiver. The elevation and depression target images (lines of sight) shall be positioned in the same vertical plane with respect to the "0" horizontal line of sight throughout 1000 mils to an accuracy of .1 of a mil. The target images shall be capable of measuring the accuracies specified in 4.6.5 when introduced into the rangefinder system at the positions corresponding to the elevation scale of the telescope and dial scale of micrometer. The target image shall be equipped with a "0" calibrated vertical line capable of measuring the accuracy of lateral displacement specified in 3.7.1.3.5. Tolerance marks shall also be incorporated on both sides of the "0" vertical line to determine plumb travel accuracies specified in 3.7.1.5.2 with respect to the "0" vertical target line.

(b) Auxiliary diopter power.- When the rangefinder system is subjected to the tests in 4.6.5 the wall target may be outside the range of eyepiece adjustment. In this case utilize an auxiliary diopter lens, between +.60 to +.65, positioned close to the sighting objective assembly of the rangefinder system in the same plane of the entrance pupil to obtain best focus of the wall target images. The eyepiece assembly may also be adjusted, if necessary, to obtain best focus setting of the target images. Upon completion of the set-up procedures the parallax shield of the rangefinder system shall be secured in the closed detent position thereby reducing the amount of stray light entering the rangefinder system and parallax which may be present at the target distance specified in 4.5.2.3.2(a).

4.6 Test methods and procedures.

4.6.1 Fabrication.

4.6.1.1 Rangefinder system components.- The inspection records of each rangefinder system submitted for final acceptance shall show evidence that the components indicated in 3.2 conform to the requirements specified therein. The voltage marking (Vo voltage value) on the cavity assembly utilized in the rangefinder shall be recorded on the inspection record of the rangefinder system along with the value of capacitance of the rangefinder system.

4.6.2 Environmental.- The rangefinder system shall be thoroughly examined for physical defects after each environmental test. A visual and tactile examination shall be made for missing parts and physical damage. Upon completion of all environmental tests, the rangefinder system in its specified configuration shall be subjected to the tests specified in 4.6.4 through 4.6.10.

4.6.2.1 Storage temperatures.- The test equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926. The test shall be performed as specified in Procedure II of MIL-F-13926 at the temperature specified in 3.5.11.

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4.6.2.2 Shock.- The test equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926. The test shall be performed as specified in Procedure II of MIL-F-13926 except the shock pulse shall comply with the wave form, time duration and peak "G" value specified in 3.5.2.

4.6.2.3 Vibration.- The test equipment utilized in this test shall conform to the vibration apparatus requirements specified in MIL-STD-810. The rangefinder system shall be firmly mounted to the vibration table. The vibration equipment shall provide the conditions of acceleration amplitude, frequency and duration specified in 3.5.1. The vibration test shall then be performed in accordance with Method 510 Procedure I, Part 1 of MIL-STD-810 at the temperatures indicated in 3.5.1.

4.6.2.4 Rough handling.- The test equipment utilized in this test shall conform to the vibration apparatus requirements specified in MIL-STD-810. The rangefinder shall be firmly mounted to the vibration table. The rough handling test shall be performed in accordance with Method 514, Procedure XI, Part 2 of MIL-STD-810 and the conditions specified in 3.5.3.

4.6.2.5 Rain.- The rain test shall be performed as outlined in MIL-F-13926, Procedure I with the following exceptions: Five (5) hours of continuous rainfall at a rate of 2 inches/hour and wind velocity of 35 knots. Temperature of the water shall be maintained at approximately 85°F. Subsequent to the rainfall test, Item 2 shall be subjected to the additional requirements specified in 3.5.4. Upon completion of the tests the requirements specified in 3.5.4 shall be met.

4.6.2.6 Submersion.- The submersion test shall be performed in accordance with Method 512, Procedure 1 of MIL-STD-810, except the temperature depth of submersion and time duration specified in 3.5.5 shall apply. Subsequent to the submersion test, remove the system from the water and wipe all exterior surfaces of system dry. Open the items specified, examine the interior and contents of the items for any evidence of water penetration to determine conformance to the requirements of 3.5.5.

4.6.2.7 Humidity.- The humidity test shall be performed as outlined in MIL-F-13926, Procedure II and under the conditions specified in 3.5.6 for four (4) hours.

4.6.2.8 Sand and dust.- The sand and dust test shall be performed as outlined in MIL-F-13926, with the following conditions. The sand and dust particle sizes, density, wind velocity, temperature, time duration and relative humidity shall be as specified in 3.5.7.

4.6.2.9 Salt spray.- The salt spray test shall be performed as outlined in MIL-STD-810, Method 509, and according to the conditions specified in 3.5.8.

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4.6.2.10 Altitude-operating.- The test equipment utilized in the test shall conform to the temperature-altitude chamber apparatus requirements specified in MIL-STD-810. The test equipment shall provide the conditions of time duration, altitude and temperature specified in 3.5.9.1. The altitude test shall be performed in accordance with Method 504, of MIL-STD-810. Upon completion of the altitude test the system shall meet the requirements of 3.5.9.1.

4.6.2.11 Altitude non-operating.- Perform this test utilizing the test methods specified in 4.6.2.10 except the conditions specified in 3.5.9.2 shall be applied. Subsequent to the altitude test the system shall meet the applicable requirements specified in 3.5.9.2.

4.6.2.12 Fungus.- One (1) system in configuration I shall be subjected to the fungus test outlined in MIL-F-13926. At the end of the 28 day incubation period, the system shall be tested as specified in 4.6.4 through 4.6. for compliance with 3.5.12 through 3.10.3. Subsequent acceptance of the system shall be by certification.

4.6.2.13 Operational temperatures.- Unless otherwise specified the test shall be performed in accordance with Procedure I of MIL-F-13926 at the extreme temperatures specified in 3.5.12 and at a room temperature between 70°F and 80°F, utilizing the test equipment indicated in 4.6.2.1. The systems shall be exposed to each specified temperature for a minimum period of 2 hours. At the end of each 2 hour period, and while still at the specified temperature, the systems shall be subjected to the tests in 4.6.4 through 4.6.10.2 to determine conformance to the applicable requirements of 3.7 through 3.10.3.

4.6.3 Reliability.- Reliability testing is required to be performed in conjunction with First Article Testing[1/]. The reliability sample shall consist of six systems with a minimum of 700 ranges per system. During those system firing tests outlined in this specification, additional firing shall be performed in order that the system reliability (para. 3) may be verified.

A record shall be kept for each sample system indicating serial number, type of environmental exposure, system configuration, number of failures (type and cause), and range data (true range and rangefinder range), as well as number of targets within reticle of rangefinder, the number of targets indicated by the target discriminator lamps and position of the target discriminator switch, type of power supply, and battery use (i.e., rangings per battery serial number).

An entry will be made into the record each time the rangefinder is fired. The data collected will be reduced and compared to the following relationship:

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$$1500 \quad \frac{2T}{2}$$

$$\chi^2 .05, \quad 2(r+1)$$

where T = total number of rangings

r = number of system failures*

χ^2 = Chi Square

2(r+1) = degrees of freedom

*System failure: Any system malfunction which renders the rangefinder inoperative or operable outside the limits of the specification.

*System interruption: Any system malfunction which does not render the rangefinder inoperable within the limits set forth in this specification but requires maintenance to correct the malfunction. Although a system interruption does not penalize the reliability, it may constitute a failure of some major subassembly for which system acceptance cannot be made until that characteristic or component which caused the system interruption has been screened. Subsequent to the reliability test, the sample system shall be refurbished and readied for delivery and shipment.

1/When this specification is part of the contract, the reliability tests associated with MIL-R-60500(MU) are not applicable.

4.6.4 Performance.- The following tests shall be performed with the rangefinder system positioned in a test chamber that conforms to the "Test Facilities" requirements of MIL-F-13926 and the extreme temperature conditions specified in 3.5.12. The assembly of system configurations shall be as specified in 3.4. The test chamber shall be provided with an optically corrected viewing window and exit window to permit viewing through the rangefinder. The chamber shall be designed to permit testing of the rangefinder as specified herein. During the tests at low and high temperature, the viewing and exit windows of the test chamber shall be frost and moisture free respectively so as not to influence (optical, mechanical, etc.) measurements for each test specified at the temperatures indicated in 3.5.12.

4.6.4.1 Ranging.- When the conditions specified in 3.7.1.1 occur, the system shall indicate the range to targets specified therein.

4.6.4.2 Ranging accuracy.- Perform this test with targets located at 230, 625, 1025, 2025, 3025, 4235, 5340, and 6450 meters with at least a

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distance of 150 meters in depth between targets. The measured distance to each target between 230 and 6450 meters shall be established within plus or minus 0.5 meters. The system shall be fired at each target utilizing appropriate setting of min-range switch, and the discriminator switch where necessary. Record the system reading to each target. The recorded range reading to each target shall conform to the requirements specified in 3.7.1.1.1.

4.6.4.3 Range resolution.- Perform this test utilizing two (2) targets. Target 1 shall be positioned at 4500 meters. Target 2 shall be located at 30 meters + 0.2 meter behind Target 1. The two targets shall be clearly visible and be within the boresight square of the system reticle, when viewed through the system eyepiece assembly. Rotate the target discriminator switch to the number "1" position. The system shall be fired to the targets. Record the range reading. Rotate the target discriminator switch to the number "2" position. The system shall again be fired to the targets. Record the range reading. The range resolution of the system shall meet the requirements specified in 3.7.1.1.2.

4.6.4.4 Target discrimination.- Perform this test with a group of five targets positioned between 250 and 550 meters. The group of targets shall be positioned as required in 3.7.1.1.3. The distance to each target shall be known. Position the min-range switch of the system to the "2" position.

1. Position discriminator switch to the "1" position and perform the following operations:

a. Adjust the orientation of the system so only the first of the five targets is visible with the boresight square and fire the rangefinder. Depress the read switch and observe the first target discriminator lamp illuminated and the range to the target.

b. Adjust the orientation of the system so that only the first and second target are visible within the boresight square and fire the system. Depress the read switch and observe only the second target discriminator lamp illuminated and the range to the first target.

c. Adjust the orientation of the system so that only the first three targets are visible within the boresight square and fire the system. Depress the read switch and observe only the third target discriminator lamp illuminated and the range to the first target.

d. Adjust the orientation of the system so that only the first four targets are visible within the boresight square and fire the system. Depress the read switch and observe only the fourth target discriminator lamp illuminated and the range to the first target.

e. Adjust the orientation of the system so that all five of the targets are visible within the boresight square and fire the system. Depress the read switch and observe only the fifth target discriminator lamp illuminated and the range to the first target.

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2. Place the target discriminator switch in the "2" position and repeat operations "1a" through "1e" above observing the same results except that the range observed shall be to the second target when it is visible with the boresight square and a no target range indication otherwise.

3. Place the target discriminator switch in the "3" position and repeat operations "1a" through "1e" above observing the same results except that the range observed shall be to the third target when it is visible within the boresight square and the no target range indication otherwise.

4. Place the target discriminator switch in the "4" position and repeat operations "1a" throught "1e" above observing the same results except that the range observed shall be to the 4th target when it is visible within the boresight square and the no target range indication otherwise.

5. Place the target discriminator switch in the "5" position and repeat operations "1a" through "1e" above observing the same results except that the range observed shall be to the fifth target when it is visible within the boresight square and a no target range indication otherwise.

4.6.4.5 Minimum range selection.- Perform this test with the targets positioned at known distances as specified in 3.7.1.1.4, Columns II and III. With the minimum range selector switch rotated to each position indicated in Column 1, and the discriminator switch positioned to the number "1", the system shall be fired at the applicable range indicated in Column II, there shall be no record of a range reading on the numerical readout. Subsequent to this test the system shall be fired to the target ranges specified in Column III. Record the range readings. Upon completion of all the ranging tests the minimum range switch shall have met the requirements specified in 3.7.1.1.4, at the indicated ranges.

4.6.4.6 Rapid ranging.- Perform this test utilizing a target positioned at a known distance of 1500 meters plus or minus 10 meters. Orient the system so that the target is clearly visible in the boresight square of the reticle. The system shall be fired at 15 second intervals for 5 consecutive firings followed by 10 consecutive firings at 1 minute intervals. The requirements specified in 3.7.1.1.5 shall be met.

4.6.4.7 Delay ranging.- Perform this test utilizing a target positioned at a known distance of 5000 ±100 meters. Orient the system so that the target is clearly visible in the boresight square of the reticle. Depress the charge switch, then after a delay of at least 75 seconds depress the range switch. The correct range to the target shall be indicated by the numerical display as specified in 3.7.1.1.6 at the temperature indicated.

4.6.4.8 Minimum time between rangings.- With the power switch in the "ON" position depress the range switch, then depress the charge switch. The time interval between the depression of the charge switch, and the audible popping

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noise of the discharging of the pulse forming network through the flash lamp shall conform to the requirements of 3.7.1.1.7, as measured with a standard stopwatch.

4.6.4.9 Self ranging.- While being subjected to the tests specified in 4.6, the system shall have not triggered unless the range switch was depressed.

4.6.4.10 Turn-on display.- Position the power switch to "ON". Depress the read switch while observing the four numerical indicators and target discriminators shall conform to the requirements specified in 3.7.1.2.1.

4.6.4.11 Time interval counter.- Position the power switch to "ON". Depress the test switch, then depress the read switch while observing the numberical display. The counter shall conform to the requirements of 3.7.1.2.2.

4.6.4.12 Range counter reset.- Position the power switch to "ON". Depress the test switch, then depress the read switch while observing the numerical display. The counter shall conform to the requirements of 3.7.1.2.3. Depress the test switch then depress the read switch while observing the numerical display. A range less than 100 meters but greater than zero shall be observed. Depress the charge switch allow ten seconds to elapse then depress and hold the read switch and then the range switch. The counter shall conform to the requirements of 3.7.1.2.3 by the display of the reset indication for approximately one second just prior to the display of the range reading.

4.6.4.13 Long range reading.- Perform this test with a target positioned at a distance greater than 9999 meters. Upon ranging on the target, the range reading on the numerical indicators shall meet the requirements of 3.7.1.2.4.

4.6.4.14 Target discriminator lamps.- Perform this test utilizing the test targets specified in 4.6.4.4.

a. The indication of the number of targets shall be displayed by the lighting of the appropriate target discriminator lamp corresponding to the multiple targets observed in the open boresight square of the system when the read switch is depressed. The requirement specified in 3.7.1.2.5a shall be met.

b. Position an illuminated 100 watt frosted incandescant lamp on the optical axis of the sight receiver, at a distance of 9 feet $\pm$ 1 foot from the sight receiver objective lens. Position the power switch to "ON". Depress the read and test switches simultaneously and observe the five neon target discriminator lamps. The five neon target discriminator lamps shall conform to the requirements specified in 3.7.1.2.5b. A standard stopwatch shall be utilized for measuring the sequencing rate of the lighting and extinguishing of the lamps.

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4.6.4.15 Low voltage indicator.- Perform this test with the input voltages to the system adjusted to the specified voltage values indicated in 3.7.1.2.6. Position the power switch to "ON". Depress the charge switch, the range switch, then the read switch while observing the numerical indicators. The number "2" shall appear in the unit column in addition to Numbers 0, 1, 5 or 7. The low voltage indicator shall conform to the requirements specified in 3.7.1.2.6.

4.6.5 Perform the following tests 4.6.5.1 through 4.6.5.6 utilizing special testing equipment conforming to 4.5.2.3.2. The level bubble of the adapter mount shall be centered by means of the adapter leveling screws prior to performing any accuracy tests of the system.

4.6.5.1 Illumination.- Actuation of the switch on the yoke assembly shall provide illumination to the elevation and azimuth scales and the circular level vial on the mounting adapter, as specified in 3.7.1.3.1. When the reticle adjustment potentiometer knob of the telescope assembly is rotated in a counterclockwise direction the reticle light shall conform to the requirements of 3.7.1.3.1.

4.6.5.2 Elevation accuracy.- Perform this test with the elevation scale of the teslecope assembly and dial scales of the micrometer of the system indicating the following elevation and depression settings in mils, +75, +150, +300, +500 "0" horizontal line of sight, -75, -150, -300, and -500 mils as specified in 3.7.1.3.2. With backlash included for the direction of travel of the elevation mechanism, utilizing the elevation knob adjust the system to zero elevation as indicated on the elevation scale of the telescope and dial scales of the micrometer with respect to their zero index without overtravel. Observe through the system eyepiece assembly, the "0" horizontal reticle line of the system shall be in coincidence with respect to the "0" horizontal line of sight of the target image. Continue rotation of the elevation knob in the same direction of travel to elevate the line of sight of the system to each mil increment specified above without overtravel. As each mil increment is established on the elevation and micrometer scales, sight through the system eyepiece assembly. The "0" horizontal reticle line of the system shall be in coincidence with horizontal target images corresponding to the elevation and depression settings introduced on the elevation and micrometer scales. Any displacement errors between the target images with respect to the "0" horizontal reticle line of the system due to elevation change between any two target points shall indicate the true position of the line of sight of the system. Record the elevation and depression accuracy readings of the line of sight at each mil increment as measured on the target image. The recorded accuracy readings shall be within the tolerances specified in 3.7.1.3.2.

NOTE: Should overtravel occur during rotation of the elevation knob, the test is nullified. The test for elevation accuracy shall be repeated.

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4.6.5.3 Elevation backlash.- Perform this test with backlash included for the direction of travel of the elevation mechanism. Adjust the system to zero elevation as indicated on the elevation scale and dial scale of the micrometer to their respective zero indix without overtravel. Observe through the eyepiece assembly of the system, the "0" horizontal reticle line of the system shall be in coincidence with respect to the "0" horizontal line of sight of the target image. Continue rotation of the elevation know in the same direction of travel to elevate the line of sight of the system approximately 25 mils, then reverse the direction of travel of the elevation knob with backlash included to return the elevation scale and dial scales to their respective zero index. View through the eyepiece assembly of the system. Any displacement error between the "0" horizontal reticle line with respect to the "0" horizontal target image line shall be recorded as measured on the target image. Backlash readings shall then be made at randomly selected target images (a minimum of 10 positions) throughout +500 mils of elevation excursion using the above test procedure for direction of travel of the elevation mechanism. Record the elevation backlash measurements. Upon completion of the elevation backlash tests, return the system to the "0" horizontal line of sight of the target image. Utilizing the above test procedures, elevation backlash reading shall be performed (a minimum of 10 positions) throughout -500 mils of depression Record the depression backlash measurements. The recorded backlash measurements shall not exceed the tolerance specified in 3.7.1.3.3.

NOTE: Should overtravel occur during the rotation of the elevation knob, the test is nullified. The test for elevation backlash shall then be repeated.

4.6.5.4 Elevation knob torque.- Standard torque measuring equipment capable of measuring the required torque values shall be utilized for this test. An appropriate adapter to accommodate the elevation knob shall be used with the torque measuring equipment. Torque shall be measured while rotating the knob counterclockwise and clockwise through the full range of required travel. The measured torque required to rotate the knob in the counterclockwise and clockwise direction shall not exceed the value specified in 3.7.1.3.4 for the temperatures indicated therein.

4.6.5.5 Lateral displacement.- Conduct this test in sequence to the test in 4.6.5.3. Elevate the system to zero elevation, as indicated on the elevation scale. Establish coincidence of the vertical reticle line with the vertical target image line. Verify the level of the system by means of the circular level vial bubble. While observing the vertical reticle line of the system, with reference to the vertical target image line elevate the system in 100 mil increments to a minimum of 500 mils elevation by means of the elevation knob. At each 100 mil increment of elevation, reverse the direction of rotation of the elevation knob by one half a revolution, then continue to the next 100 mil increment of elevation. Any horizontal movement of the vertical reticle line of the system as a result of this elevation procedure, shall not exceed the tolerances specified in 3.7.1.3.5 as measured on the target.

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Return the system to zero elevation. Depress the line of sight of the system in 100 mil increments for a total of 500 mils utilizing the above procedure. During this 500 mil excursion there shall be no horizontal displacement of the vertical reticle line in excess of the tolerances specified in 3.7.1.3.5 as measured on the vertical target image line.

4.6.5.6 Plumb travel.- Conduct this test in sequence to the test in 4.6.5.5. Elevate the system 500 mils as indicated on the elevation scale. Establish coincidence of the vertical reticle line of the system with the vertical target image line. Verify the level of the system by means of the circular level vial bubble. Depress the line of sight of the system the required number of mils, by means of the elevation knob, while observing the vertical reticle line with reference to the vertical target image line. During this excursion the vertical reticle line of the system shall track the target image line within the tolerance specified in 3.7.1.5.2, as measured on the target.

4.6.6 Azimuth excursion.- Rotate the adapter mount spindle to its zero limit stop and slip the azimuth scale of the mount to indicate zero mils in azimuth with respect to its index. View through the microscope of the system and adjust the microscope eyepiece for best focus of the adapter mount azimuth scale. Rotate (Slew) the transceiver through the excursion specified in 3.7.1.4.1 while viewing through the microscope. The azimuth excursion of the adapter mount shall meet the requirements specified in 3.7.1.4.1 as measured on the azimuth scale of the adapter mount and microscope.

4.6.6.1 Azimuth error.- Perform this test utilizing the zero alignment procedure specified in 4.6.6 prior to performing the azimuth error test. The azimuth readings shall be made at 10 randomly selected calibrated target images in each quadrant. While observing through the microscope rotate the system by hand for (course adjustment) to each selected target, then rotate the azimuth knob of the adapter mount for fine adjustment to the target image. Record each azimuth reading as measured on the azimuth scale of the adapter mount and microscope. Upon reaching the limit stop return the adapter mount to its azimuth zero index stop. Continue testing the remainder of the selected azimuth target images as noted above. The total recorded azimuth readings to the selected target images shall not exceed the tolerances specified in 3.7.1.4.2 as measured on the azimuth scale of the adapter mount and microscope.

4.6.6.2 Azimuth knob rotation.- The maximum torque required to rotate the azimuth knob adjustment for azimuth rotation of the transceiver shall be measured by means of standard torque measuring equipment and appropriate adapter. The measured torque to rotate the azimuth knob shall not exceed the values specified in 3.7.1.4.3 at the temperature indicated.

4.6.6.3 Microscope.- Perform this test utilizing a calibrated dioptometer with a magnification of at least 3 power (3X) for measuring the accuracy of the

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microscope focus as specified in 3.7.1.5.3. Set the dioptometer objective scale to zero and adjust the dioptometer eyepiece for best focus of the dioptometer reticle. Set the microscope eyepiece assembly to its mid-point of travel. Position the dioptometer at the microscope eyepiece assembly. Observe through the dioptometer, obtain best focus of the microscope reticle by adjustment of the dioptometer objective scale. Record the diopter value as measured on the dioptometer objective scale. Obtain best focus of the azimuth scale by adjustment of the dioptometer objective scale. Record the diopter value as measured on the dioptometer objective scale. The difference between the two recorded dioptometer scale values shall not exceed the tolerance specified in 3.7.1.4.4.

4.6.6.4 Circular level.- The level bubble of the adapter mount shall conform to the requirements specified in 3.7.1.5.1 as the transceiver is rotated through the azimuth excursion specified in 3.7.1.5.1.

4.6.6.5 Open sight.- A calibrated wall target capable of measuring the open sight accuracies of the system assembly as specified in 3.7.1.6.1 shall be utilized with this test. Construction of the wall target shall contain an established level line of sight and a plumb line. While viewing through the telescope eyepiece assembly the telescope vertical and horizontal reticle lines shall be in coincidence with respect to the plumb line and established level line of sight. A calibrated auxiliary telescope of at least three power (3X) containing a vertical and horizontal reticle line shall be positioned on a leveled surface of the testing equipment at the open sight. The horizontal reticle line of the telescope shall be aligned to the horizontal (top) machined surface of the open sight with the vertical reticle line bisecting the vertex of the "V". Observe through the auxiliary telescope and record the position of the intersecting point of the reticle lines with respect to the target image utilized to test the open sight. The intersecting point of the reticle lines shall not exceed the tolerance specified in 3.7.1.6.1. as measured on the target.

4.6.6.6 Collimation (Boresight).- Adjust the syteem to its zero azimuth position as indicated by the microscope reticle and mounting adapter scale. Adjust the system to zero elevation as indicated on the elevation scale of the telescope assembly and dial scale of the elevation knob. Cross-level the system by means of the mounting adapter leveling screws and circular level vial. Position the Boresight Chart: 10541240 as specified in 3.7.1.6.2 with its boresight cross aligned to the system reticle. Set the min-range switch to the ".2" position, and the target discriminator switch to the "1" position. Position the power switch to "ON". Fire the laser beam at boresight chart, while adjusting the system in azimuth and elevation, until the laser beam is centered in the rectangle pattern of the target. When the laser beam is centered in the rectangular pattern of the boresight target, the horizontal and vertical reticle line of the system, the elevation and dial scales of the transcriver shall conform to the requirements of 3.7.1.6.2. Close the parallax shield. The position of the open cross of the system reticle, with reference to the hole in the parallax shield shall conform to the requirements of 3.7.1.6.2.

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4.6.6.7 Boresight adjustment.- Standard torque measuring equipment capable of measuring the required torque values shall be utilized for this test. An appropriate adapter to accommodate the boresight screws shall be used with the torque measuring equipment. The measured torque to rotate the azimuth and elevation shall not exceed the values specified in 3.7.1.6.3 at the temperatures indicated.

4.6.6.8 Eyepiece focus.- The test shall be made with the aid of a calibrated dioptometer with a magnification of at least 3 power (3X) and a diopter range for measuring the accuracies of the diopter knob scale as specified in 3.7.1.6.4. Adjust the dioptometer eyepiece for best focus of the dioptometer reticle and set the dioptometer objective scale to zero. Set the diopter scale of the telescope to zero diopter. Position the dioptometer at the telescope assembly eyepiece. While viewing through the dioptometer, obtain best focus of the telescope assembly reticle by adjustment of the dioptometer objective scale. The telescope reticle shall be in best focus within the limit specified in 3.7.1.6.4, as measured on the dioptometer objective scale.

4.6.6.9 Diopter adjustment.- Standard torque measuring equipment capable of measuring the required torque values shall be utilized for this test. With an adapter affixed to the diopter knob, rotate the knob through the range specified in 3.7.1.6.5. The maximum torque shall not exceed the values specified in 3.7.1.6.5 at the temperatures indicated.

4.6.6.10 Reticle illumination.- Apply the operating voltage to the reticle lamp of the telescope as specified in 3.7.1.6.6. The illuminated reticle etchings shall be clearly defined when visually examined against a darkened background. Should the brightness of a reticle defect be greater than the reticle lines, the defect shall be cause for rejection. The potentiometer knob shall be rotated clockwise from the extreme counterclockwise position. During this rotation the potentiometer shall meet the requirements specified in 3.7.1.6.6 while viewing the reticle pattern.

4.6.7 Battery power.- The battery power shall be capable of meeting the requirements specified in 3.7.2.1 at the temperatures indicated.

4.6.7.1 Charger function.- The system in Configurations C, D and E shall meet the requirements of 3.7.1 through 3.7.1.6.6 at the temperatures indicated in 3.7.2.2a. The system in configuration C or D, item 5, shall meet the requirements of 3.7.1 through 3.7.1.6.6 and charging time at the temperatures indicated in 3.7.2.2b. Subsequent to charging, the system shall be subjected to the requirements specified in 3.7.2.1.

4.6.7.2 Mounting stability.- The system in Configurations A through E and Configuration J shall meet the requirements specified in 3.7.3.

4.6.7.3 Tripod.- The legs of the tripod (item 3) shall slide without interference when the tripod is adjusted from minimum to maximum height. With the legs of the tripod extended, the requirements specified in 3.7.3.1 shall be met.

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4.6.7.4 Shelf mount.- The stakes of the shelf mount (item 12) shall be capable of being inserted into the angular hole in the base as specified in 3.7.3.2.

4.6.7.5 Weight.- The weight of individual items of the system shall not exceed the tolerances specified in 3.7.4.1.

4.6.7.5.1 Operability.- The interlocking fasteners cable connectors, hinges, shoulder strap buckles and mechanical strap fasteners shall meet the requirements of 3.7.4.2 and function freely without restriction or binding when subjected to a visual and tactile examination.

4.6.8 Electromagnetic stability.- The system in Configuration A through E shall be subjected to test methods of MIL-STD-462 for Class IIA equipment for compliance with 3.8.

4.6.9 Interchangeability.

4.6.9.1 Selection.- Three items shall be selected as follows from initial production to determine conformance with 3.9:

a. Prior to final assembly, a group of 3 of each of the 14 items listed in 3.9 shall be presented to the Quality Assurance Representative for selection and final assembly of 3 systems.

b. System No. 1 shall be assembled by randomly selecting 1 item from each group of 3 of the 14 grouped.

c. System No. 2 shall be likewise assembled by randomly selecting 1 item of each of the 2 remaining in each of the 14 groups.

d. The remaining 14 items will be assembled as system No. 3.

4.6.9.2 System.- The 3 systems obtained from 4.4.1.2 shall be subjected to all requirements and tests of this specification in compliance with 3.9. The selection (4.4.1.2), final assembly and examination and tests shall be witnessed by the Quality Assurance Representative. Subsequent to Government approval and acceptance of the first 6 systems, 3 additional systems shall be selected, assembled and submitted for interchangeability tests in accordance with the above procedures at 25%, 50% and 75% completion of the contract production and witnessed by the Quality Assurance Representative each time. When the manufacturing process, materials, or inspection techniques previously applied are subject to change, or an adverse change occurs in performance capability or reliability of the system, this shall be cause to cease those operations contributing to the change or failure until corrections have been made by the contractor and approved by the Government.

4.6.9.3 Locking torque.- Perform this test utilizing standard torque measuring equipment equipped with an appropriate adapter to accommodate the assembly under test. Torque shall be measured while rotating the locking

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knob of the adapter mount (item 2) through the maximum clockwise excursion, to secure the transceiver as specified in 3.9a. The measured torque value shall meet the requirements specified in 3.9a.

4.6.9.4 Unlocking torque.- Perform this test in sequence with the test specified in 4.6.9.3. Torque shall be measured while rotating the locking knob of the adapter mount through the excursion to unlock and release the transceiver as specified in 3.9b. The measured torque value shall meet the requirements specified in 3.9b.

4.6.9.5 Support interface.- The carrying cases of Configurations F and Chest of Configuration I shall interface with and support intended items as specified in 3.9c(1). In Configuration J the universal tripod shall be capable of seating the adapter mount (with transceiver) and securing it in place. The requirements specified in 3.9c(2) shall be met.

4.6.10 Operation and maintenance.- Conduct this test in the following sequence:

a. Position the power switch to "ON". Momentarily depress the charge switch and allow a ten second time period for the pulse forming network to charge. Immediately following the ten second time period position the power switch to "OFF". Subsequent to these operations, the pulse forming network and motor of the cavity assembly shall conform to the requirements of 3.10.1a, as determined by an audible examination. Discharging of the pulse forming network through the flash lamp will be recognized as the distinctive popping noise that occurs when the transceiver is fired.

b. Position the power switch to "ON". Depress the range switch without depressing the charge switch. Subsequent to this operation the motor of the cavity assembly shall conform to the requirements of 3.10.1b, as determined by an audible examination.

c. Turn the switch of the transceiver to the "ON" position. Press the charge switch to charge the capacitors of the PFNPS. Leaving the switch in the "ON" position, remove the cover of the left housing. (Immediately turn switch to "OFF" position after cover is removed). The capacitors shall have discharged through the load resistor located in the left housing as indicated by a voltmeter reading of no more than 50 volts when a voltmeter probe is placed on the contact spring of the bracket assembly (10540971) subsequent to separating the spring from contact by an insulator. The requirements of 3.10.1c shall be met.

4.6.10.1 Eye filter.- The filter level of the eye filter assembly shall meet the requirements specified in 3.7.10.2(b) when subjected to all environmental conditions as specified in 3.10.2b. The filter assembly shall be removed and replaced (a minimum of 5 times) from the transceiver line of sight to determine repeatability positioning of the filter assembly and operability of the filter lever.

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4.6.10.2 Equipment protection.- The system shall meet the requirements specified below:

a. The adapter mount shall prevent 360° rotation of the transceiver as specified in 3.10.3a.

b. The charger shall prevent damage to separate internal functions within the charger as specified in 3.10.3b.

c. The adapter mount cable shall prevent damage to the transceiver caused by short circuit within the transceiver as specified in 3.10.3.c.

d. The cable connectors items 15, 16 and 17 shall prevent connection to a wrong receptacle within the system.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking.- Packaging, packing, and marking shall be in accordance with Packaging Data Sheet 10541244 for Rangefinder System, Laser, AN/GVS-3. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- This instrument is used by the forward observer and provides precise target location data in the form of azimuth, elevation, and range (mils for azimuth and elevation; meters for range).

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, date, and number of this specification.
b. Order of document precedence.
c. Applicable packaging data sheet number.
d. Selection of applicable levels of preservation, and packing.
e. Provisions for First Article Testing.

6.3 Assembly of system configurations.- Assemble items 1 thru 17 of para. 3.4 into configurations A through J specified in 3.4 as required by 6.3.1 through 6.3.10.

6.3.1 Configuration A.- Figure 1 shows configuration A with transceiver and battery pack mounted on the tripod and adapter mount. Assemble these items into configuration A as follows:

a. Extend the legs of the tripod and coarse level the tripod.

b. Place the adapter mount on the tripod as shown in Fig. 2. Lock the adapter mount onto the tripod by turning the locking lever.

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c. Place a transceiver with yoke assembly on the adapter mount. Lock the yoke to the adapter mount by rotating the locking and contact knob.

d. Place the battery pack on the tripod leg and strap it securely in position.

e. Connect the adapter mount connector plug to the recepticle at the top of the battery pack.

6.3.2 Configuration B.- Fig. 3 shows configuration B with the transceiver, battery, adapter mount and shelf mount assembled. Assemble these items into configuration C as follows:

a. Place the adapter mount on the shelf mount as shown in Fig 2 and secure the adapter mount to the shelf mount by turning the locking lever.

b. Place the transceiver with yoke assembly on the adapter mount and secure the yoke assembly to the adapter mount by rotating the locking and contact knob.

c. Connect the adapter mount connector plug to the plug at the top of the battery pack.

6.3.3 Configuration C.- Configuration C is the same as configuration A except that the battery charger shall be used in place of the battery pack.

a. Connect the rangefinder end of the 50' cable (10541236) to the adapter mount cable connector.

b. Connect the other end of the 50' cable (10541236) to "Output for Rangefinder" connector located on the battery charger (see Fig. 8).

c. Connect the 12' cable (10541237) to "Input 115 VAC" of the battery charger (see Fig. 8).

d. Connect the other end of 12' cable (10541236) to a 110 volt, 400 cycle per sec, AC power supply.

6.3.4 Configuration D.- The assembly of configuration D is the same as configuration C except that the other end of the 12' cable (10541236) shall be connected to a 110 volt, 60 cycle per sec, AC power supply as opposed to a 110 volt, 400 cycle per sec, AC power supply.

6.3.5 Configuration E.-

a. Assemble items 1, 2, 3, 15 and 17 as indicated in configuration B except that cable 10541237 shall be replaced by cable 10544852 as indicated below.

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b. Secure the plug end of cable 10544852 to the "28-Volt D.C. Input" connector of the battery charger as indicated in Fig. 4.

c. At the other end of the cable separate enough of the two cable wires, which have spade lugs, loosen the nuts on the battery connectors and slip the spade lugs under the nuts, red wire on the positive side and black wire on the negative side. Secure nuts firmly.

6.3.6 Configuration F.- Store items 1, 2, 3, 4 and 12 into carrying cases, items 6 and 7 as shown in Fig. 5 and as described below:

a. Place rangefinder with yoke assembly into carrying case, 10541038.

b. Place folded tripod with adapter mount and battery pack attached into carrying case, 10541026.

c. Place shelf mount into carrying case, 10541026.

d. Close carrying cases and attach as shown in Fig. 6.

6.3.7 Configuration G.- Place three battery packs in the battery charger as shown in Fig. 7. Connect the appropriate connectors of the battery charger to the plug at the top of each battery pack.

6.3.8 Configuration H.- Store items 6 and 7 with contents (see 6.3.6) and item 9, into the chest (item 8) as shown in Fig. 7.

6.3.9 Configuration I.- Store items 1 through 18 excluding item 16 and 14 in appropriate containers as indicated in Figs. 7 and 8. Install containers in a housing simulating a jeep, M 151, according to the following procedures:

a. Chest Installation

(1) With the chest locked in the brackets, position the chest on the vehicle floor, centrally located between the fender wells and 5 inches from the inside rear wall of the vehicle. Mark the vehicle floor through the 8-holes in the brackets.

(2) Remove chest from the vehicle floor. Center drill in each of the 8-marks on the M151 vehicle floor. Enlarge the center drilled holes to 13/32 inch diameter.

(3) Install bolts through the two holes in each of the brackets and the M151 vehicle floor, and secure loosely with plates, lock washers and nuts under M151 vehicle floor.

(4) Reinstall chest on brackets and insert removable pins into forward brackets to lock chest to floor of M151 Vehicle. Tighten bolts firmly under M151 vehicle floor.

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(5) Remove slack that may be between the chest and brackets by adjusting the set screws located in the back of the rear brackets.

b. Cable Holder Installation.

(1) Position the cable holder against the side of the right fender well 2 inches above the floor and 4 inches from the back wall of the vehicle. Mark the fender well through the six holes in the cable holder.

(2) Remove the cable holder and drill each of the marks thru with a 9/32" dia. drill.

(3) Position the cable holder against the inside of the fender wall and insert 6 hex head screws 1/4-20NCx1/2 inch long through cable holder and fender wall. Secure loosely from underside of fender well, using lock washers and nuts. Align cable holder then secure nuts and bolts firmly.

c. Battery Charger Installation.

(1) Position the battery charger on the right rear fender well 13 inches $\pm$ 1/2 inch from the inside rear of the M151 vehicle and flush with the left side of the fender well.

(2) Using the battery charger as a template, mark the fender well thru the six holes in the battery charger. Remove the battery charger and drill a 7/16 in. dia. hole in each of the marks in the fender well.

(3) Position one long and one short plate on top of the fender well with the short plate 10544821 forward of the long plate-10544824. Align holes through fender well with holes through plate weld nuts. Using the plates as a jig, drill three (F) .257 diameter holes through the fender well. Remove the plates and enlarge the holes to 7/16 inch dia.

NOTE: Bolts may be installed through the underside of the fender well and into the seld nuts to secure the plates to aid the jig drilling.

(4) Reposition the plates on the underside of the fender well and secure with a hex head screw and lock washer in the outboard hole. Place battery charger on top of fender well and realign with 7/16 dia. hole and tapped hole in weld nuts. Insert hex head bolts, washers and secure firmly.

6.3.10 Configuration J.- Figure 4 show configuration J with the range-finder and battery pack mounted on an adapter mount and universal tripod. Assemble these items as follows:

a. Place the adapter mount on the tripod and lock the adapter mount to the tripod by turning the screw located in the head of the tripod.

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b. Place the rangefinder with yoke assembly on the adapter mount and secure the yoke to the adapter mount by rotating the locking and contact knob.

c. Hook the battery pack on the crossbar of the tripod leg.

d. Connect the adapter mount connector plug to the plug at the top of the battery pack.

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Preparing Activity:

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Project No. 1240-A247

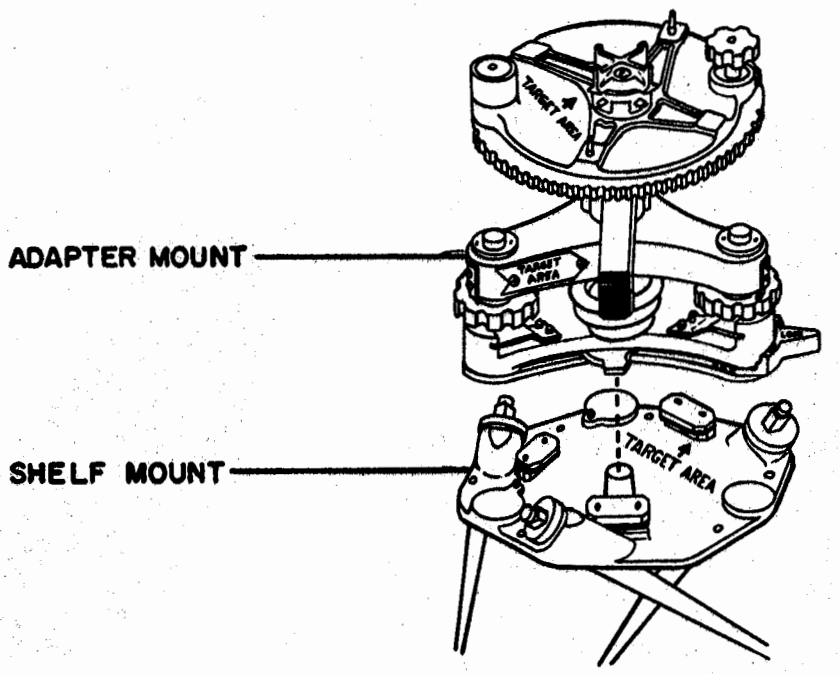
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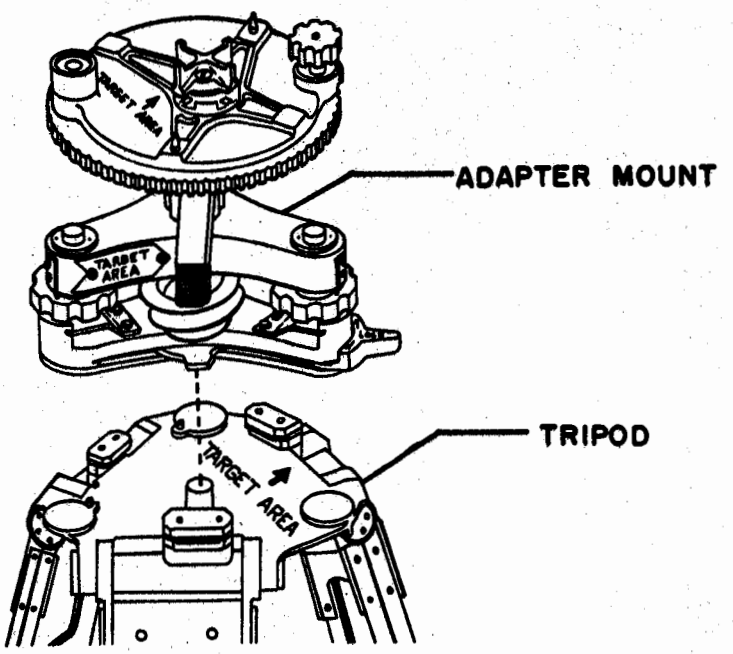


CONFIGURATION A - RANGE FINDER AN/GVS-3 MOUNTED ON TRIPOD

FIGURE 1.

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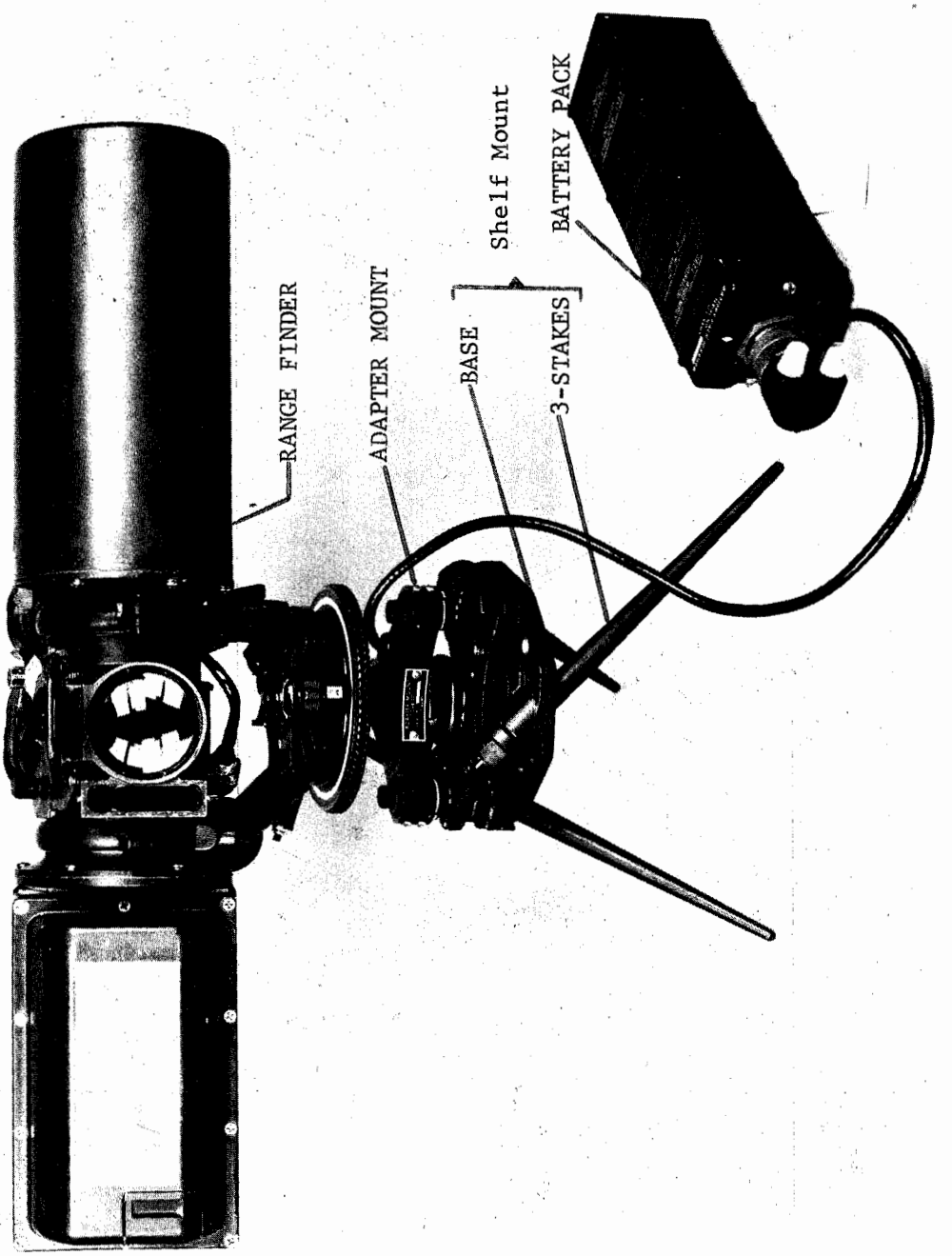




ALINEMENT OF TARGET AREAS FOR PROPER INSTALLATION

FIGURE 2.

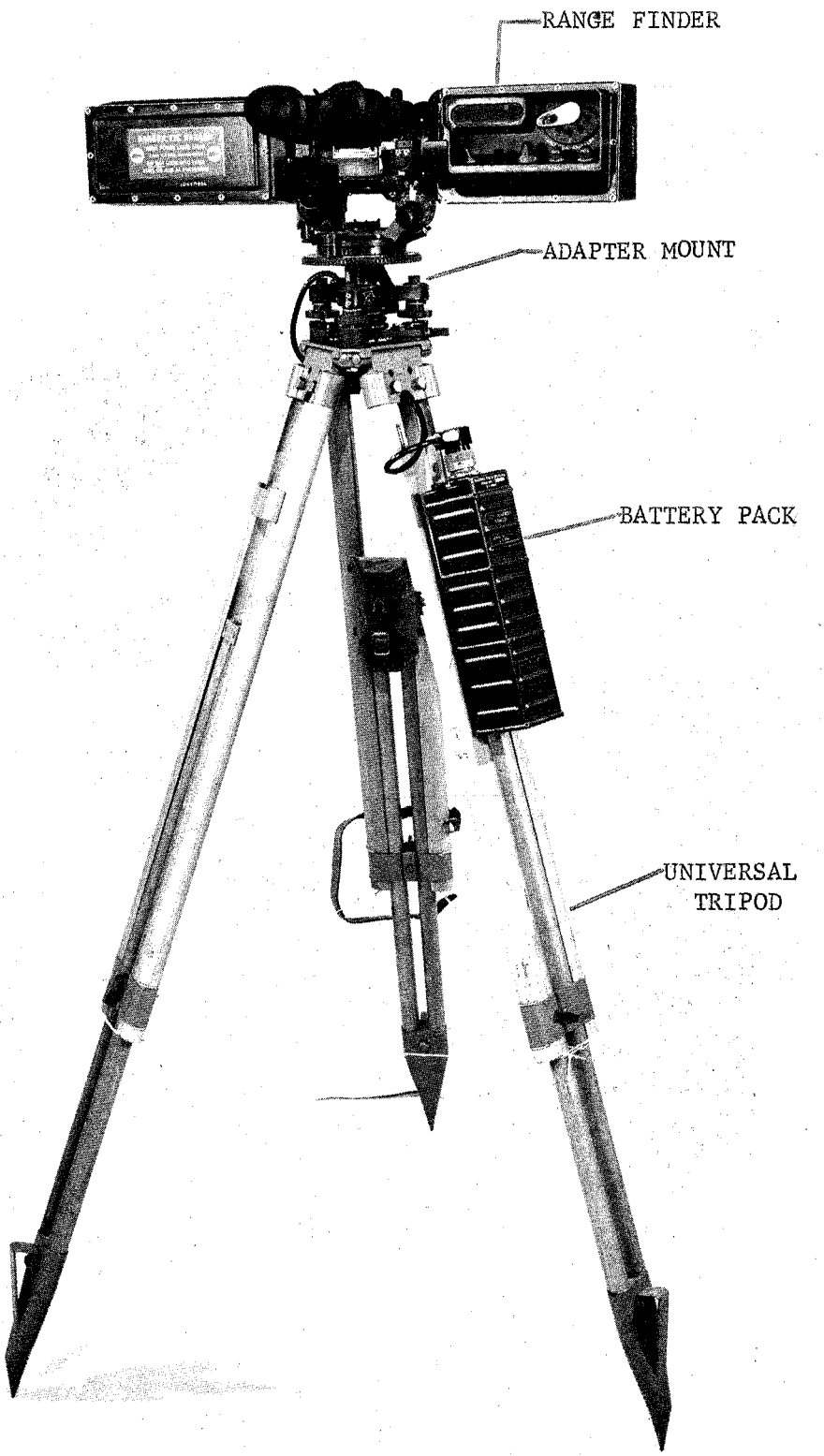
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CONFIGURATION C - RANGE FINDER MOUNTED ON SHELF MOUNT

FIGURE 3.

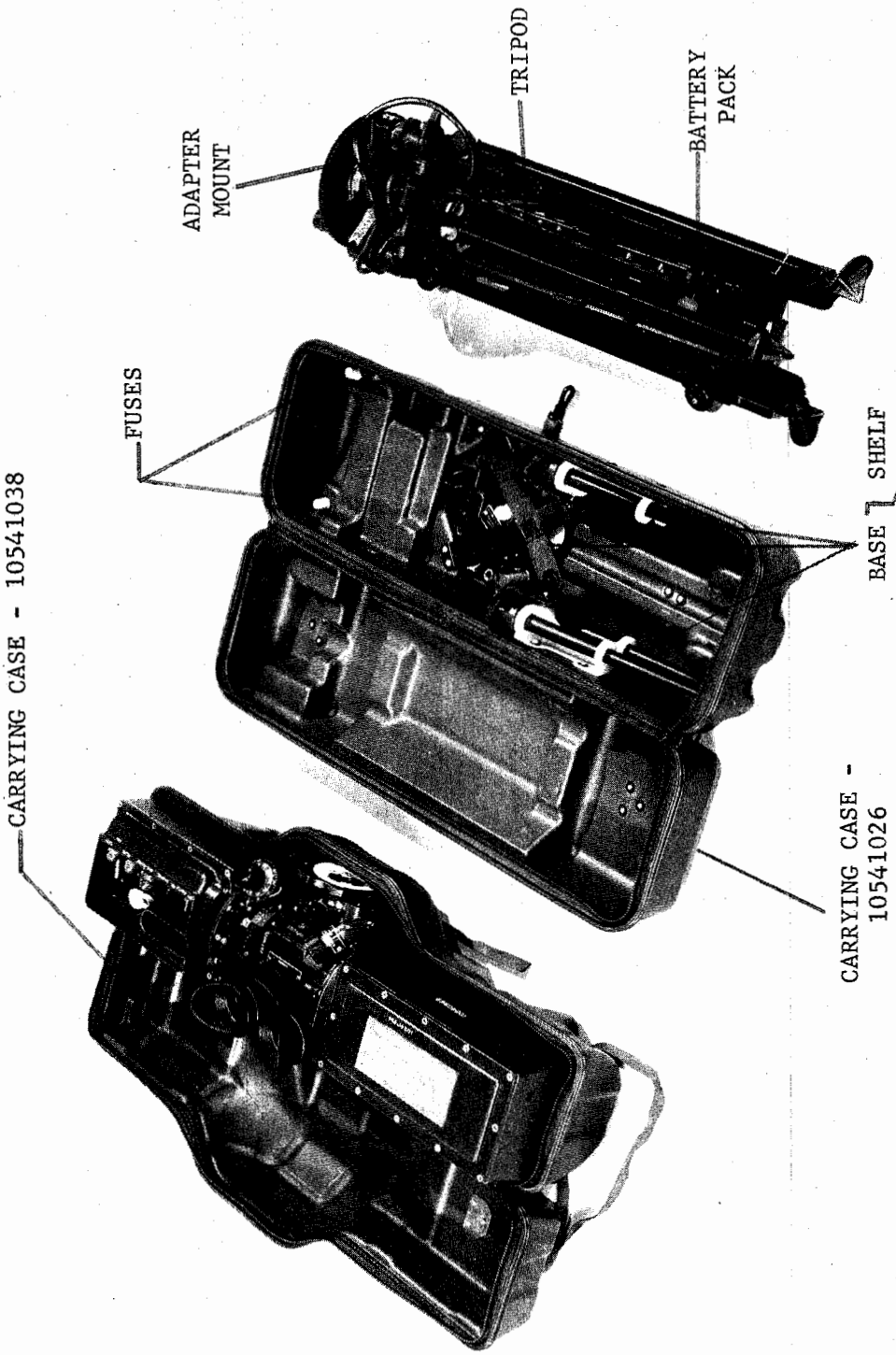
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CONFIGURATION E - RANGE FINDER AN/GVS-3 MOUNTED ON UNIVERSAL TRIPOD

FIGURE 4.

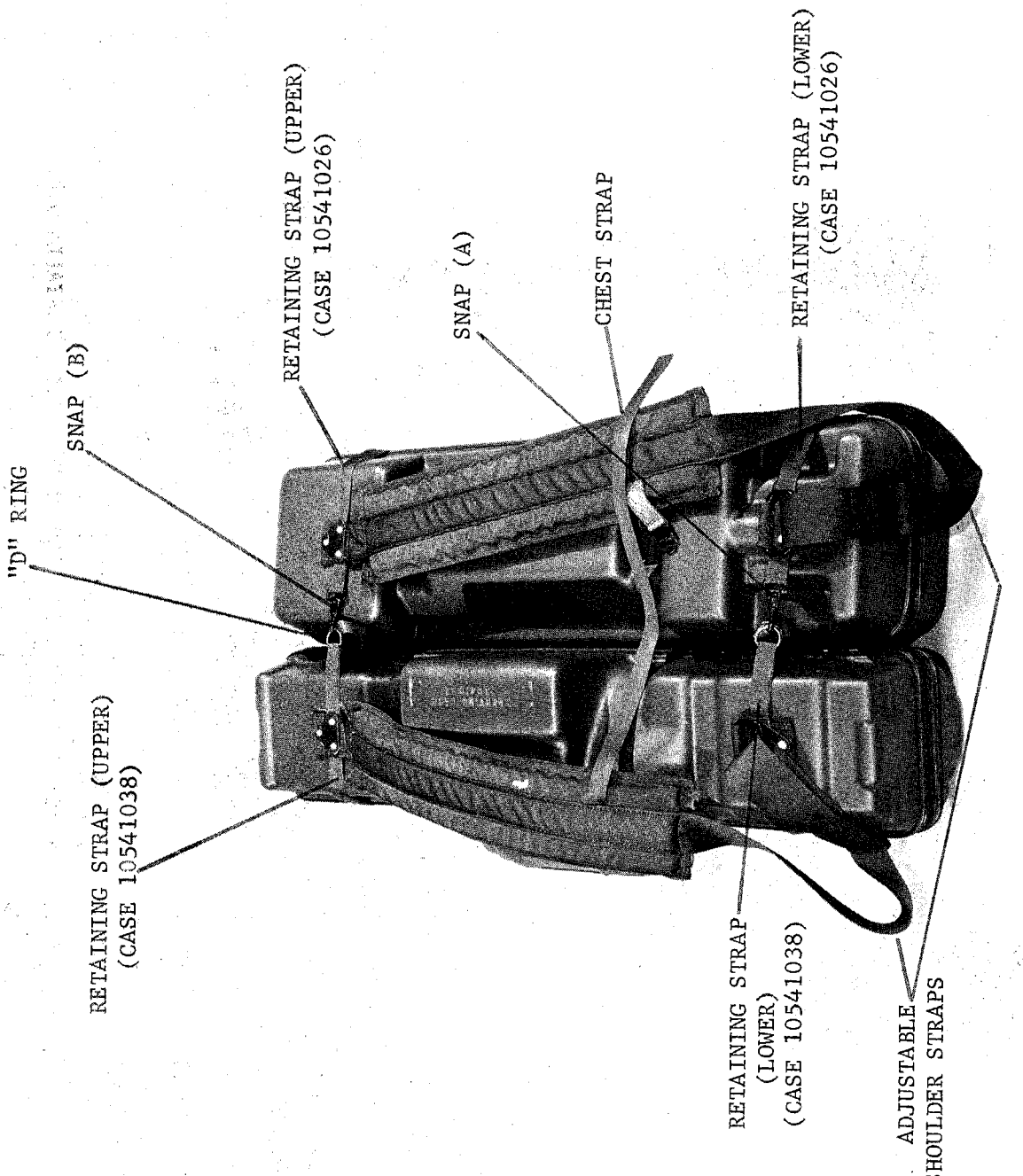
Source: https://assist.dla.mil - Downloaded: 2026-10-26T04:02Z
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CONFIGURATION F - ACCESSORY EQUIPMENT (BACKPACK)

FIGURE 5.

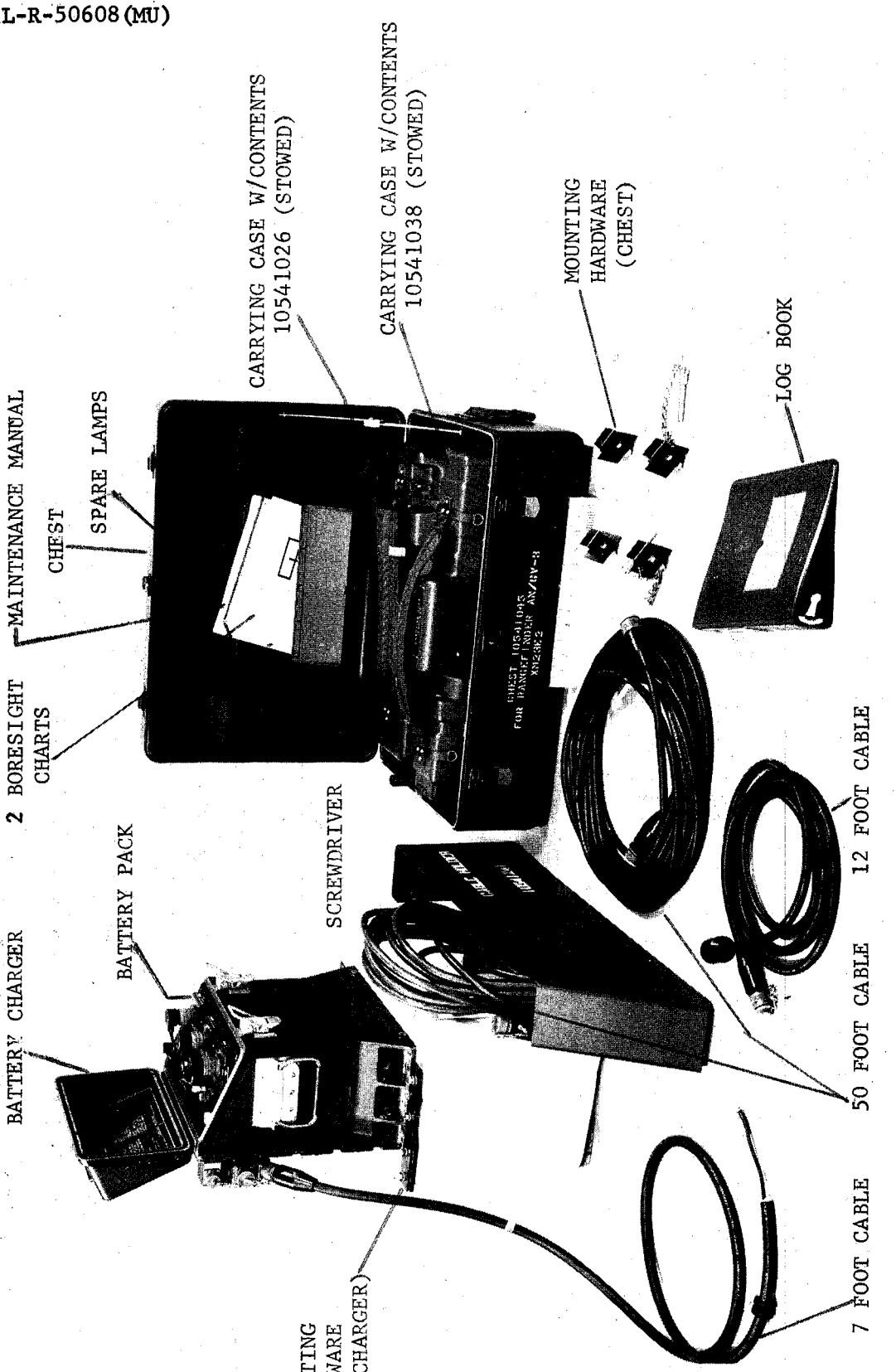
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:02Z
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CONFIGURATION F - CARRYING CASES CONNECTED FOR BACKPACK

FIGURE 6.

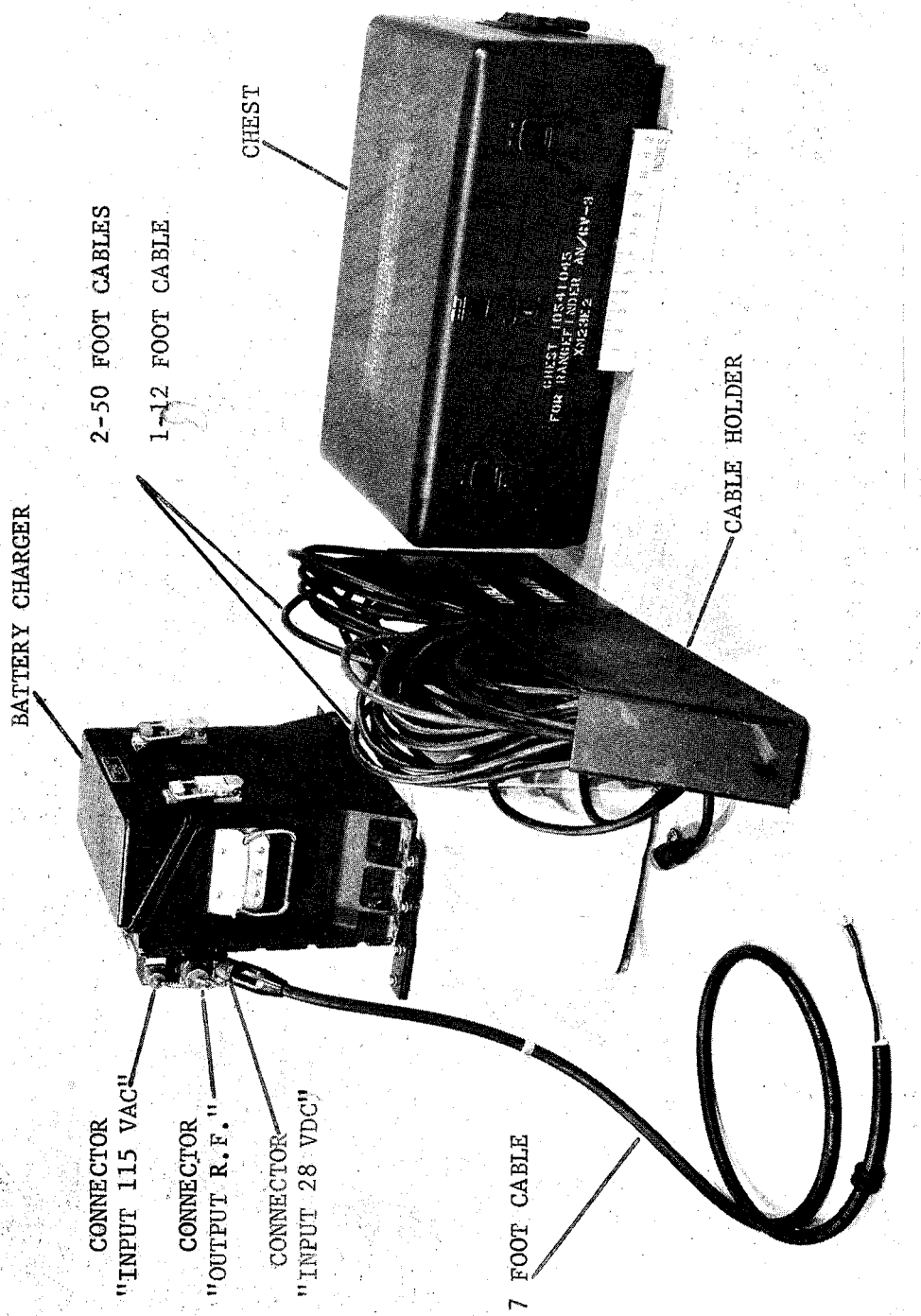
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:02Z
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CONFIGURATION I - ACCESSORY EQUIPMENT (JEEP INSTALLATION)

FIGURE 7.

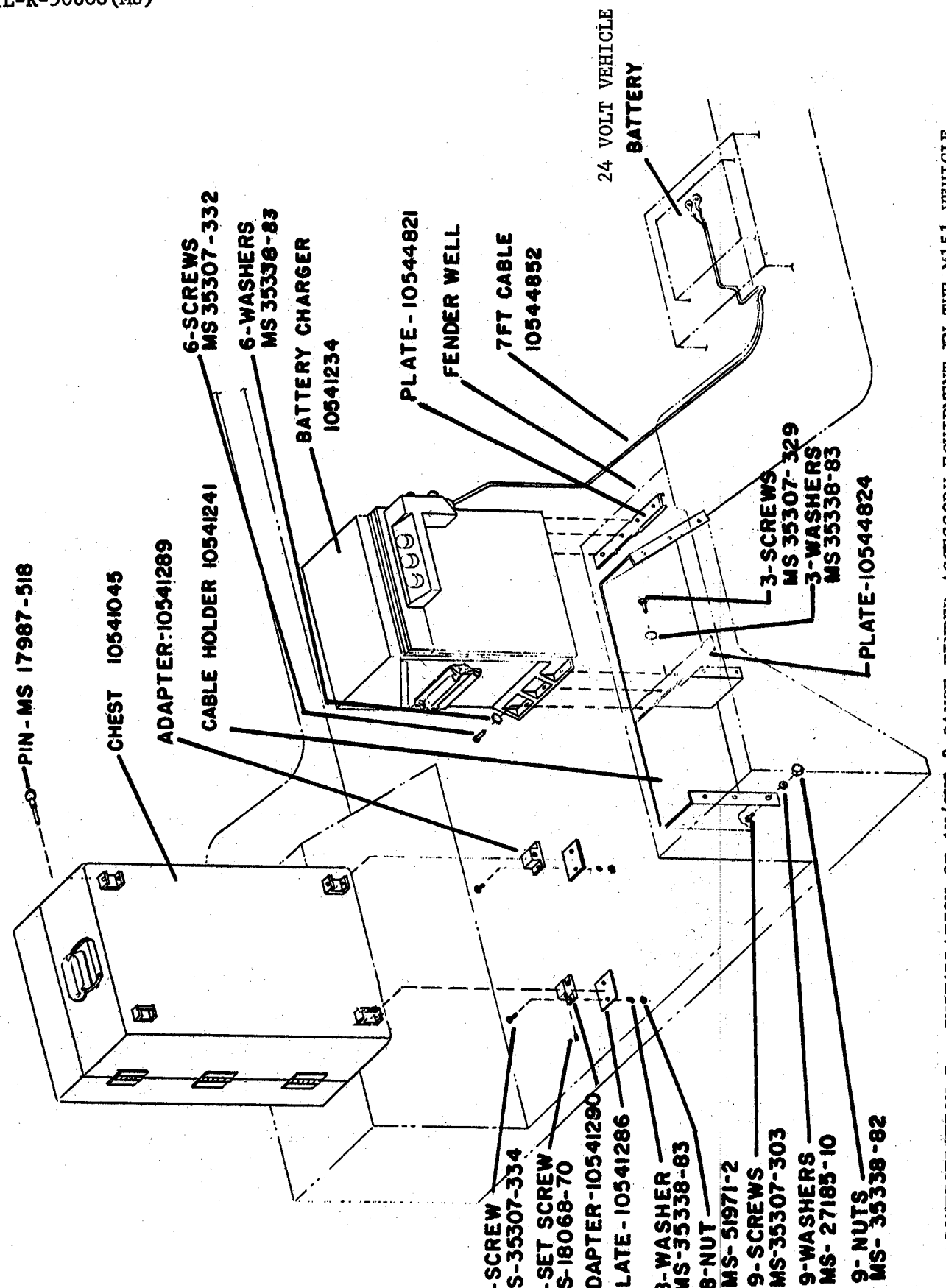
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CONFIGURATION I - ACCESSORY EQUIPMENT STOWED (JEEP INSTALLATION)

FIGURE 8.

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CONFIGURATION I - INSTALLATION OF AN/GVS-3 RANGE FINDER ACCESSORY EQUIPMENT IN THE M151 VEHICLE
FIGURE 9.

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